

Problem

Design an op amp circuit to perform the following operation:

$$v_o = 3.5v_1 - 2.5v_2$$

All resistances must be $\leq 100 \text{ k}\Omega$.

Step-by-step solution

Step 1 of 5

The required output voltage equation is,

$$v_o = 3.5v_1 - 2.5v_2 \dots\dots (1)$$

Use cascade form of one inverting amplifier and one summing amplifier to design the required output voltage equation. The circuit is shown in Figure 1.

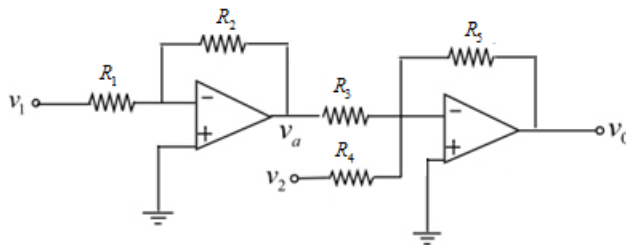


Figure 1

Step 2 of 5

Refer to Figure 1 and write output voltage equation for inverting amplifier.

$$v_a = -\left(\frac{R_2}{R_1}\right)v_1$$

Refer to Figure 1 and write output voltage equation for summing amplifier.

$$v_o = -R_5\left(\frac{v_a}{R_3} + \frac{v_2}{R_4}\right)$$

$$v_o = -\frac{R_5}{R_3}v_a - \frac{R_5}{R_4}v_2$$

Substitute $-\left(\frac{R_2}{R_1}\right)v_1$ for v_a .

$$v_o = -\frac{R_5}{R_3}\left(-\frac{R_2}{R_1}\right)v_1 - \frac{R_5}{R_4}v_2$$

$$v_o = \left(\frac{R_2R_5}{R_1R_3}\right)v_1 - \frac{R_5}{R_4}v_2 \dots\dots (2)$$

Step 3 of 5

Compare first terms of equations (1) and (2).

$$\frac{R_2 R_3}{R_1 R_3} = 3.5$$

Let us choose feedback resistors, $R_2 = R_3$ as $100\text{ k}\Omega$ and substitute in the equation.

$$\frac{(100 \times 10^3)(100 \times 10^3)}{R_1 R_3} = 3.5$$

$$3.5 R_1 R_3 = 10 \times 10^9$$

$$R_1 R_3 = 2.857 \times 10^9$$

$$R_3 = \frac{2.857 \times 10^9}{R_1}$$

Choose input resistor, R_1 as $100\text{ k}\Omega$ and substitute in the equation.

$$R_3 = \frac{2.857 \times 10^9}{100 \times 10^3}$$

$$R_3 = 28.57\text{ k}\Omega$$

Step 4 of 5

Compare second terms of equations (1) and (2).

$$\frac{R_5}{R_4} = 2.5$$

$$R_4 = \frac{R_5}{2.5}$$

Substitute $100\text{ k}\Omega$ for R_5 .

$$R_4 = \frac{100 \times 10^3}{2.5}$$

$$R_4 = 40\text{ k}\Omega$$

Step 5 of 5

Therefore the designed resistor values are:

$$R_1 = 100\text{ k}\Omega,$$

$$R_2 = 100\text{ k}\Omega,$$

$$R_3 = 28.57\text{ k}\Omega,$$

$$R_4 = 40\text{ k}\Omega,$$

$$R_5 = 100\text{ k}\Omega.$$

The designed circuit is shown in Figure 2.

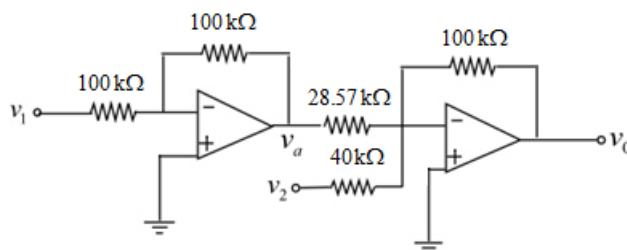


Figure 2